

Energy Relations for Electron and Hole States

A step-by-step study guide to your boss's slide and how it maps to the $\chi^{(2)}$ / Nextnano calculation

Core idea: The slide is explaining a practical simplification: calculate the confined-state wavefunctions and optical matrix elements at $k_{\parallel} = 0$, keep the energies k -dependent, then integrate the energy-dependent response over $k_{\parallel}$.

Energy relations for electron and hole state

$$\chi_{ijk}^{(2)} = \frac{N_z e^3}{2\epsilon_0 \hbar} \sum_P \sum_{b1, b2, b3} \sum_{l, m, n} f_{b,l} \int \frac{\langle \phi_{1,l}^* | r_i | \phi_{2,m} \rangle \langle \phi_{2,m}^* | r_j | \phi_{3,n} \rangle \langle \phi_{3,n}^* | r_k | \phi_{1,l} \rangle}{[E_{b2,m}(k_{\parallel}) - E_{b1,l}(k_{\parallel}) - E_1 - E_2][E_{b3,n}(k_{\parallel}) - E_{b1,l}(k_{\parallel}) - E_2]} 2\pi k_{\parallel} dk_{\parallel}$$

Calculated at $k_{\parallel}=0$, zone center can come out of integral

Energies of light for frequency doubling $E_1=E_2$

k•p calculation using Optel, Nextnano or NRL Multibands

1. Get Energy Solutions for confined direction and associated wavefunctions, ϕ .
2. Calculate $\langle \phi_{1,l}^* | r_i | \phi_{2,m} \rangle, \langle \phi_{2,m}^* | r_j | \phi_{3,n} \rangle, \langle \phi_{3,n}^* | r_k | \phi_{1,l} \rangle$
3. Get the energy of the solutions in the free motion directions
4. Fit the energy dependence.
5. Assuming free electron like with a constant effective mass these terms should look like

$$E_{b,m}(k_{\parallel}) = \frac{\hbar^2 k_{\parallel}^2}{2m_{b,m}^*} + E_{b,m}(k_{\parallel} = 0)$$

6. Note $E_{b,m}(k_{\parallel} = 0)$ is the energy solution in the confined direction

Source figure: 'Energy relations for electron and hole state' supplied by the user.

1. The slide in plain English

One-minute version: Solve the quantum well once at the zone center to get $E_n(0)$ and the wavefunctions. Use those wavefunctions to calculate the numerator. Then describe how each state energy bends with k_{parallel} , use those $E_n(k)$ values in the resonance denominators, and integrate over k .

$$\chi^{(2)} \sim M(0) \int_0^{k_{\text{max}}} \frac{2\pi k_{\parallel} dk_{\parallel}}{D_1(k_{\parallel})D_2(k_{\parallel})}$$

The equation is being split into two jobs

Part of equation	What it means	How the slide treats it
Numerator: matrix elements	How strongly the three quantum states couple optically.	Calculate at $k_{\text{parallel}} = 0$ and, as an approximation, treat as constant over the integral.
Denominators: energies	How close each optical pathway is to resonance.	Keep $E_{b,m}(k_{\text{parallel}})$ explicitly dependent on k_{parallel} .
$2\pi k_{\text{parallel}} dk_{\text{parallel}}$	Radial weighting of in-plane k states.	Integrate all relevant in-plane momentum states.

What the yellow highlight means: The yellow highlight on the slide marks the part your boss wants to calculate at $k_{\text{parallel}} = 0$ and move outside the integral.

2. What does $k_{\text{parallel}} = 0$ mean?

Definition: $k_{\text{parallel}} = 0$ is the in-plane zone center. The carrier has no in-plane kinetic contribution there, so the energy is the confined-state solution $E_{b,m}(0)$.

- **Confined direction z:** the heterostructure creates discrete states such as e1, e2, hh1, and hh2.
- **Free in-plane directions x-y:** the same confined state can carry different in-plane momentum k_{parallel} .
- **At $k_{\text{parallel}} = 0$:** you are at the bottom/reference point of the simple subband dispersion used on the slide.

The state label now has two parts

confined state (e1, e2, hh1, hh2) + in-plane momentum k_{parallel}

Important conceptual jump: An e1 subband is not one isolated energy when in-plane motion is included. It is a function $E_{e1}(k_{\text{parallel}})$.

3. Step 1 - solve the confined quantum-well problem

The first bullet on the slide says to get the energy solutions in the confined direction and the associated wavefunctions. For your workflow, this is the role of a band-structure / k.p solver such as Nextnano.

1. Define the layer structure: material compositions, well widths, barrier widths, and interface profiles.
2. Solve the quantum problem in the growth direction z .
3. Obtain confined-state energies at the zone center, such as $E_{e1}(0)$, $E_{e2}(0)$, $E_{hh1}(0)$, and $E_{hh2}(0)$.
4. Obtain the corresponding wavefunctions $\phi_{e1}(z)$, $\phi_{e2}(z)$, $\phi_{hh1}(z)$, and $\phi_{hh2}(z)$.

Slide bullet 6: $E_{b,m}(k_{\text{parallel}} = 0)$ is the energy solution of the confined problem. It becomes the reference energy in the parabolic dispersion used later on the slide.

4. Step 2 - calculate the optical matrix elements

$$M = \langle \phi_1 | r_i | \phi_2 \rangle \langle \phi_2 | r_j | \phi_3 \rangle \langle \phi_3 | r_k | \phi_1 \rangle$$

These three factors form the numerator of the general nonlinear-optics pathway. They answer: how strongly do state 1, state 2, and state 3 connect through the three dipole interactions?

- **Large matrix-element product:** the pathway can contribute strongly.
- **Small or zero matrix element:** that pathway is weak or forbidden.
- **Wavefunctions matter:** the values depend on the spatial and band character of the quantum states.

Your boss's simplification

$$M(k_{\parallel}) \approx M(0)$$

Instead of recalculating the wavefunctions and matrix elements at every $k_{\parallel}$, the slide proposes evaluating them at the zone center and using those values throughout the k integral.

What should eventually be checked: This is an approximation, not an identity. It is reasonable only if the matrix elements do not change enough over the important k range to alter the result significantly.

5. Steps 3-5 - describe how the energies change with $k_{\parallel}$

$$E_{b,m}(k_{\parallel}) = E_{b,m}(0) + \frac{\hbar^2 k_{\parallel}^2}{2m_{b,m}^*}$$

The slide assumes a simple free-electron-like, constant-effective-mass dispersion. Each confined state has a base energy plus an in-plane kinetic-energy term.

Piece	Meaning
$E_{b,m}(0)$	Energy of band b , confined state m at the zone center; this is the confinement solution.
$\hbar^2 k_{\parallel}^2 / (2m^*)$	Additional energy associated with motion in the free in-plane directions.
$m_{b,m}^*$	Effective mass controlling the curvature of that particular subband.

Why the formula is useful: This turns the Nextnano $k = 0$ energy into the bottom/reference of an $E(k_{\parallel})$ curve.

6. Effective mass: what does m^* control?

$$\frac{d^2E}{dk^2} = \frac{\hbar^2}{m^*}$$

The effective mass is a compact way of describing band curvature. It is not necessarily the bare free-electron mass.

- **Small effective mass:** larger curvature; the state energy changes quickly with k .
- **Large effective mass:** flatter dispersion; the energy changes more slowly with k .
- **Different states:** e1, e2, hh1, and hh2 can have different effective masses and therefore different dispersions.

Step 4 interpretation: The slide says to "fit the energy dependence." In practice, you can obtain $E(k)$ points from the electronic-structure calculation and test how well a parabola with a constant m^* describes the relevant k range.

7. Why the energy dependence belongs in the denominator

$$\Delta E_{nm}(k_{\parallel}) = E_{e,n}(k_{\parallel}) - E_{hh,m}(k_{\parallel})$$

The optical transition energy is the difference between the electron and hole state energies at the same $k_{\parallel}$. Because both state energies disperse with k , the transition energy generally changes with k too.

For frequency doubling / SHG

$$E_1 = E_2 = E_{\text{photon}}, \quad E_1 + E_2 = 2E_{\text{photon}} \approx 1.60 \text{ eV at } 1550 \text{ nm}$$

The first denominator is effectively asking whether a transition energy is close to the combined energy of the two input photons. At 1550 nm that combined photon energy is approximately 1.60 eV.

This is the physical reason for the integral: As $k_{\parallel}$ changes, $\Delta E(k_{\parallel})$ changes, so each k state can sit at a different distance from resonance. That is why the denominator must stay inside the k integral.

8. The complete Nextnano -> $\chi^{(2)}$ workflow implied by the slide

1. Define the coupled-QW geometry and material profile.
2. Use Nextnano (or another suitable band solver) to obtain the confined-state energies and wavefunctions.
3. At $k_{\text{parallel}} = 0$, identify the relevant e1/e2 and hh1/hh2 states.
4. Use the zone-center wavefunctions to calculate the required optical / envelope matrix elements.
5. Obtain or fit $E_{b,m}(k_{\text{parallel}})$ for the relevant states.
6. Build transition energies such as $\Delta E_{nm}(k) = E_{e,n}(k) - E_{hh,m}(k)$.
7. Insert those k-dependent transition energies into the resonance denominators.
8. Use the zone-center matrix-element product as the numerator if the approximation is being used.
9. Integrate the response over k_{parallel} with the radial factor $2\pi k_{\text{parallel}} dk_{\text{parallel}}$.
10. Apply the remaining prefactors and unit conventions to obtain $\chi^{(2)}$.

$$\chi^{(2)} \sim M(0) \int_0^{k_{\max}} \frac{2\pi k_{\parallel} dk_{\parallel}}{D_1(k_{\parallel})D_2(k_{\parallel})}$$

Computational interpretation: The expensive quantum-state information is being separated from the simpler k-dependent dispersion model. That is what makes the slide computationally attractive.

9. How interface grading enters this workflow

Changing an abrupt GaAs/AlGaAs interface to a graded interface changes the confinement potential $V(z)$. That propagates through almost every quantity on the slide.

Interface change	What changes next	How χ^2 can be affected
Composition profile changes	Band-edge / confinement potential	Changes the bound-state problem.
Bound-state problem changes	$E_n(0)$	Moves transition energies and therefore resonance positions.
Bound-state problem changes	$\phi_n(z)$	Changes optical overlaps and z-directed matrix elements.
Dispersion changes	$E_n(k)$ and potentially m^*	Changes how the resonance is sampled over k_{parallel} .

Causal chain: interface grading $\rightarrow V(z) \rightarrow E_n$ and $\phi_n \rightarrow$ matrix elements + transition energies $\rightarrow k$ integral $\rightarrow \chi^2$

10. Two assumptions to keep separate from the exact equation

Assumption A - matrix elements are constant in k

$$M(k_{\parallel}) \approx M(0)$$

This is the yellow-highlighted approximation on the slide. A more complete treatment would allow the wavefunctions and therefore the matrix elements to vary with $k_{\parallel}$.

Assumption B - the dispersion is parabolic with constant effective mass

$$E_{b,m}(k_{\parallel}) = E_{b,m}(0) + \frac{\hbar^2 k_{\parallel}^2}{2m_{b,m}^*}$$

The parabolic form is a model for the free in-plane motion. If the calculated band is noticeably non-parabolic over the important k range, a fitted $E(k)$ curve or direct k -resolved energies would be more faithful than a single constant m^* .

Hole-energy convention

For heavy-hole / valence states, check the energy convention used by your actual Nextnano output. A valence-band electron energy and a positive hole quasiparticle energy are not plotted with the same sign convention, so the dispersion formula should be applied consistently with the solver output.

Main lesson remains: These cautions do not change the main lesson of the slide: use $k = 0$ to define the confined states and matrix elements, while retaining k -dependent state energies in the optical resonance calculation.

11. Final cheat sheet

Symbol / idea	What to say in your head
$E_{b,m}(0)$	Confined-state energy at zone center.
$\phi_m(z)$	Wavefunction of that confined state.
matrix-element product M	How strongly this nonlinear optical pathway couples.
$M(k)$ approx $M(0)$	Boss approximation: calculate numerator once at zone center.
$E_{b,m}(k)$	Energy of the same state after adding in-plane motion.
m^*	How curved that $E(k)$ dispersion is.
$\Delta E_{nm}(k)$	Electron-hole optical transition energy at this k .
E_1, E_2	Input photon energies; equal for SHG.
$2\pi k dk$	Radial weighting used to add all in-plane k states.

Mastery test: If you can explain why the numerator is evaluated at $k = 0$ but the denominators remain k -dependent, you understand the central purpose of this slide.

The whole slide in one line

Nextnano at $k = 0 \rightarrow E_n(0)$, $\phi_n \rightarrow$ matrix elements + fit $E_n(k) \rightarrow k$ -dependent resonance denominators $\rightarrow$ integrate over $k \rightarrow \chi^{(2)}$

Source note

This guide is based on the boss slide supplied in the conversation and the step-by-step explanation developed directly from that slide. The cautions about the k -independent matrix-element approximation, parabolic dispersion, and hole-energy convention are explanatory implementation notes, not additional claims printed on the slide.